Year 9 Mathematics

Unit 2: Area and Volume

2.1 Surface Area of Prisms

Surface Area of Prisms

STARTER · 5 MIN

Date: _____ **Title: Surface Area of Prisms**

LEARNING OBJECTIVES

- Find the surface area of cuboids and cubes
- Find the surface area of triangular prisms
- Find the surface area of other prisms

STARTER *Answer in your book. No calculators.*

1. Area of a rectangle 7 cm by 4 cm
2. Area of a triangle: base 6 cm, height 5 cm
3. Area of a parallelogram: base 8 cm, height 3 cm
4. How many faces does a triangular prism have?

Starter — Answers

STARTER

Tick or correct in green pen

1. **28 cm²**
2. **15 cm²**
3. **24 cm²**
4. **5 faces**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the surface area of cuboids and cubes	☐
SILVER	I can find the surface area of triangular prisms	☐
GOLD	I can find the surface area of other prisms	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Surface Area of Prisms

INTRODUCTION · 3 MIN

A prism has the same cross-section all along its length.

Net

Unfold the faces flat to see them all.

A cuboid has 6 rectangular faces.

Surface area

Add the areas of all the faces.

Units: cm^2, m^2

Cuboid

Opposite faces are the same.

$SA = 2(lw + lh + wh)$

Triangular prism

2 triangles and 3 rectangles.

$$\text{triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

Examples 1 and 2 — Cuboids and Cubes

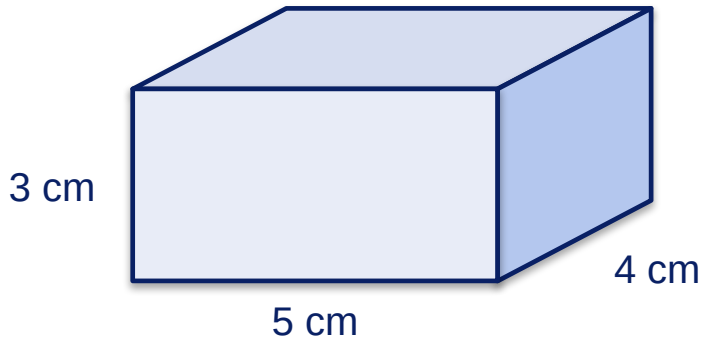
EXAMPLES · 5 MIN

BRONZE *Watch and copy into your book*

Hint SA = 2(lw + lh + wh) cube: SA = 6 × edge^2

Example 1

Find the surface area.



Example 2

A cube has edges 6 cm.

Find its surface area.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

3 pairs of faces

faces: 5 × 3, 5 × 4, 3 × 4

2 × (15 + 20 + 12)

= 94 cm²

Tip: every face has a matching opposite face.

Example 2

6 equal square faces

one face: 6 × 6 = 36

6 × 36

= 216 cm²

Tip: area units are squared.

Task 1

PRACTICE · 9 MIN

BRONZE *Find the surface area*

Hint SA = 2(lw + lh + wh) cube: SA = 6 × edge²

1. Cuboid 4 cm by 3 cm by 2 cm
2. Cuboid 10 cm by 5 cm by 2 cm
3. Cube with edges 5 cm
4. Cuboid 8 cm by 6 cm by 3 cm
5. Cube with edges 2.5 cm

★ Cuboid 7 cm by 4 cm by 1.5 cm

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **52 cm²**
2. **160 cm²**
3. **150 cm²**
4. **180 cm²**
5. **37.5 cm²**

★ **89 cm²**

Examples 3 and 4 — Triangular Prisms

EXAMPLES · 5 MIN

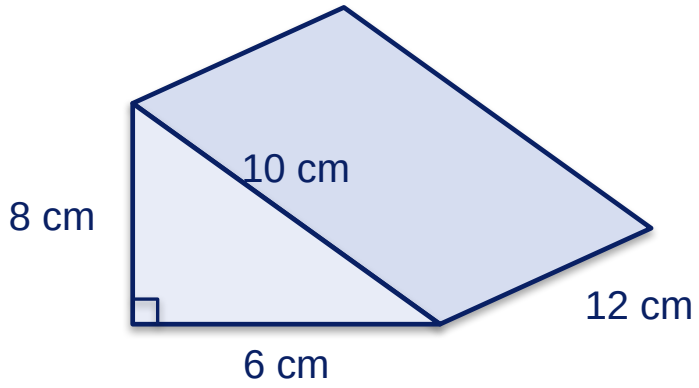
SILVER

Watch and copy into your book

Hint triangle = $\frac{1}{2}$ × base × height

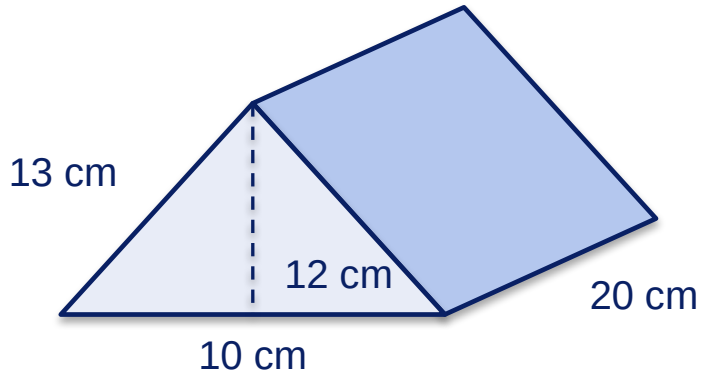
Example 3

Find the surface area.



Example 4

Find the surface area.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

2 triangles + 3 rectangles

ends: $2 \times \frac{1}{2} \times 6 \times 8 = 48$

$6 \times 12 + 8 \times 12 + 10 \times 12 = 288$

$48 + 288 = 336$ cm²

Tip: each rectangle is a side × 12 cm.

Example 4

Two sloping faces are the same

ends: $2 \times \frac{1}{2} \times 10 \times 12 = 120$

$10 \times 20 + 2 \times 13 \times 20 = 720$

$120 + 720 = 840$ cm²

Tip: 12 cm is the height, not a face edge.

Task 2

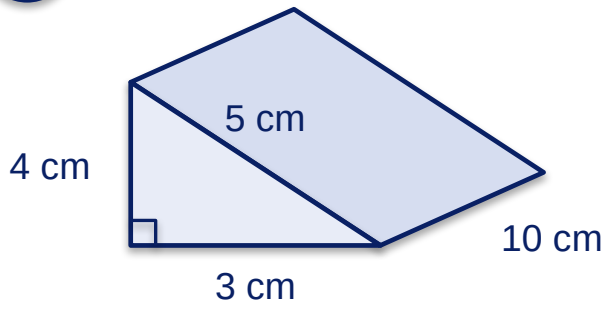
PRACTICE · 9 MIN

SILVER

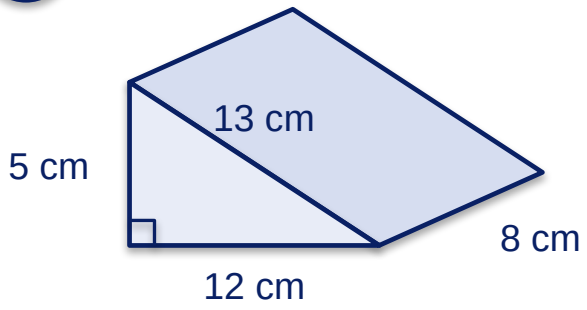
Find the surface area

Hint triangle = $\frac{1}{2}$ × base × height

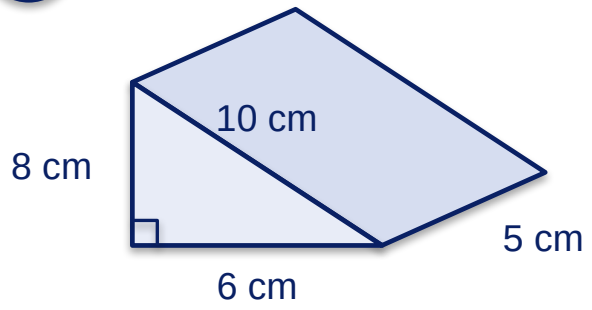
1 Surface area



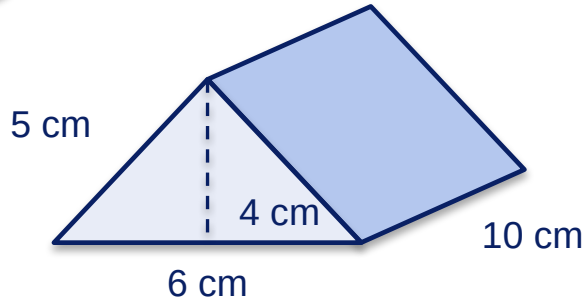
2 Surface area



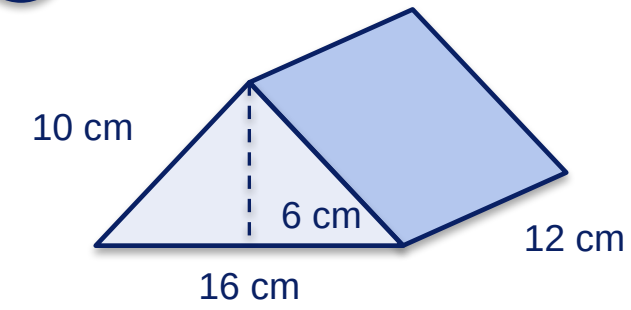
3 Surface area



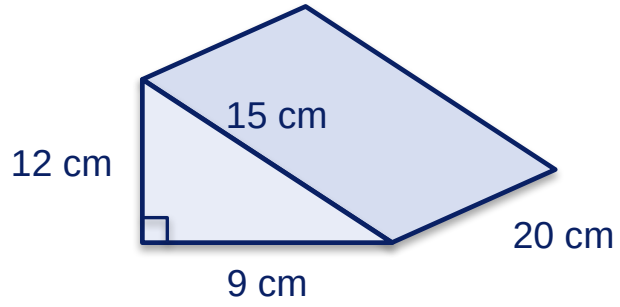
4 Surface area



5 Surface area



★ Surface area



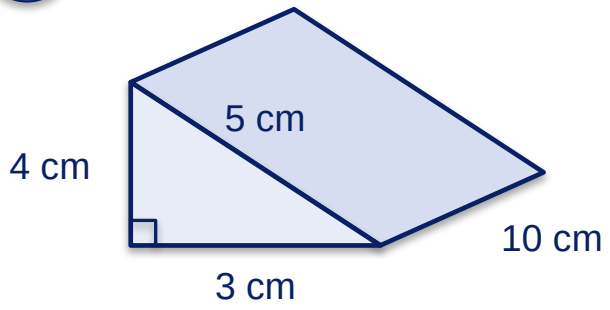
Task 2 — Answers

CORRECTIONS

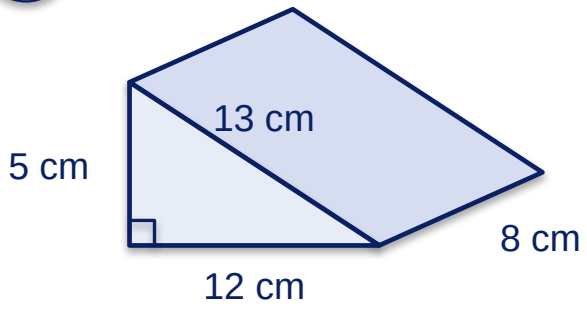
SILVER

Tick or correct in green pen

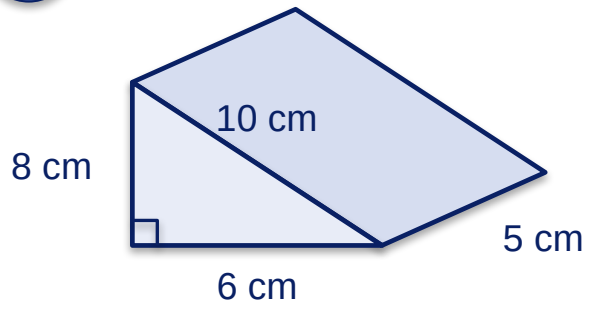
1 **132 cm²**



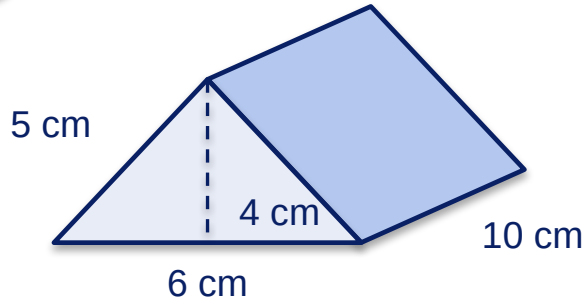
2 **300 cm²**



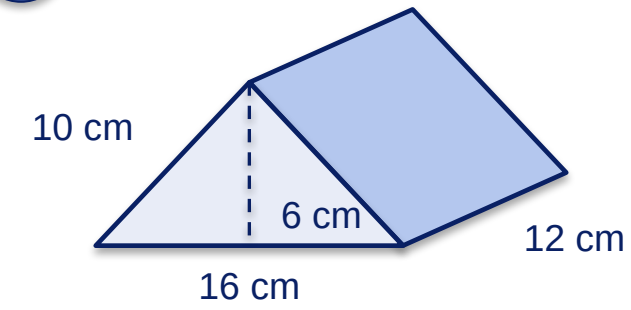
3 **168 cm²**



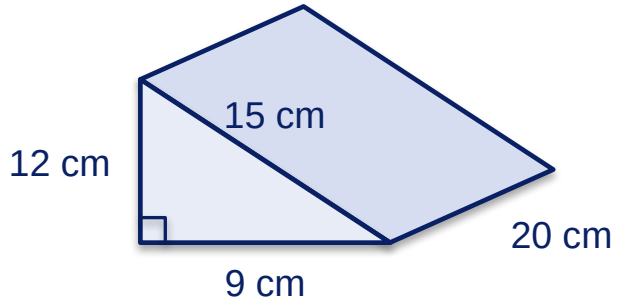
4 **184 cm²**



5 **528 cm²**



★ **828 cm²**



Examples 5 and 6 — Other Prisms

EXAMPLES · 5 MIN

GOLD

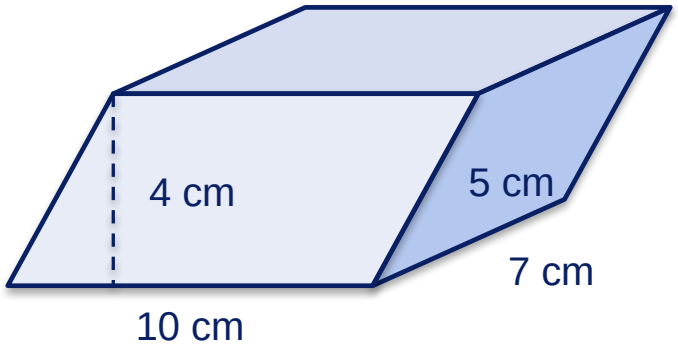
Exam-style: show all your working

Hint parallelogram = base × height trapezium = $\frac{1}{2}(a + b)h$

Example 5

(3 marks)

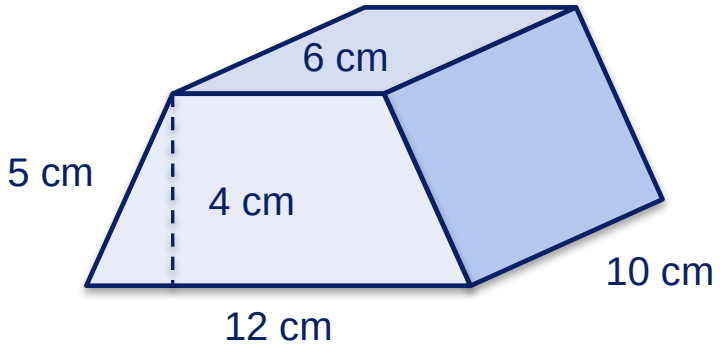
Find the surface area.



Example 6

(3 marks)

Find the surface area.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Ends are parallelograms: b × h

ends: $2 \times 10 \times 4 = 80$

$2 \times 10 \times 7 + 2 \times 5 \times 7 = 210$

$80 + 210 = 290$ cm²

Tip: use the slant side 5 cm for the faces.

Example 6

Ends are trapeziums

ends: $2 \times \frac{1}{2} \times (6 + 12) \times 4 = 72$

$60 + 120 + 2 \times 50 = 280$

$72 + 280 = 352$ cm²

Tip: list all 6 faces before adding.

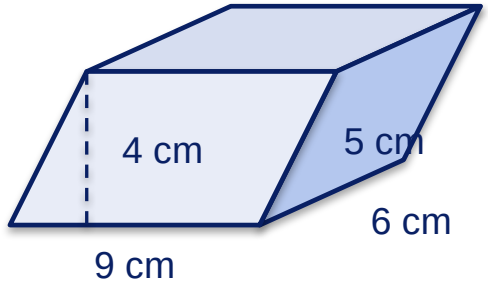
Task 3

PRACTICE · 9 MIN

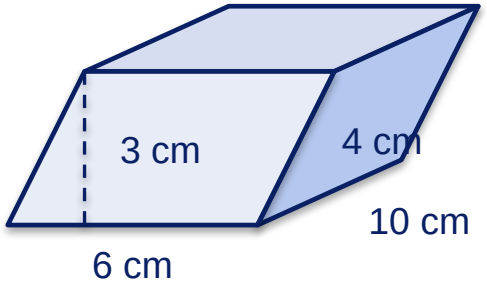
GOLD *Find the surface area*

Hint parallelogram = base × height trapezium = $\frac{1}{2}(a + b)h$

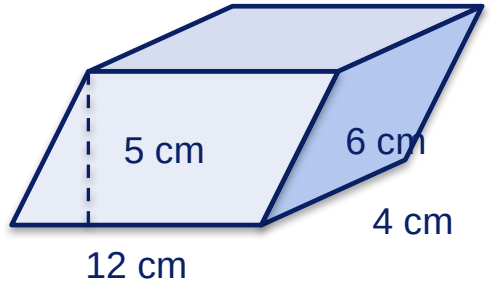
1 Surface area



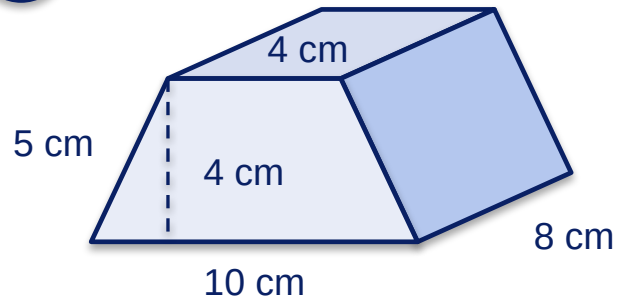
2 Surface area



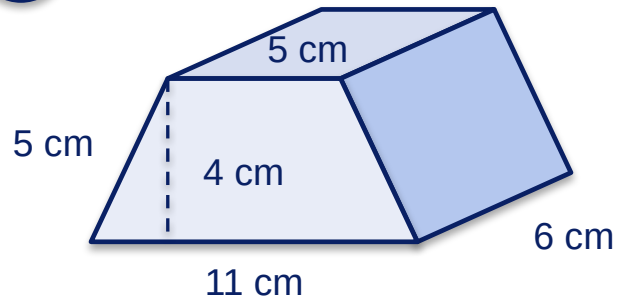
3 Surface area



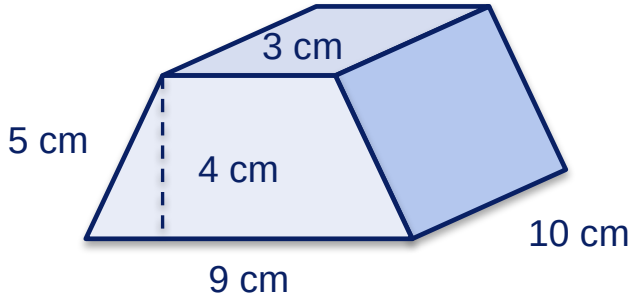
4 Surface area



5 Surface area



★ Surface area



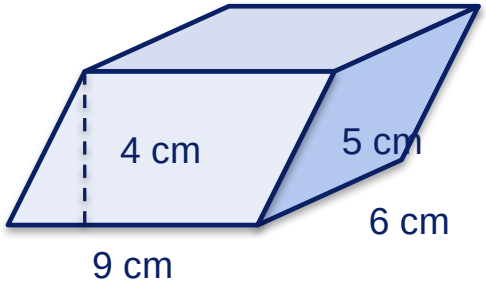
Task 3 — Answers

CORRECTIONS

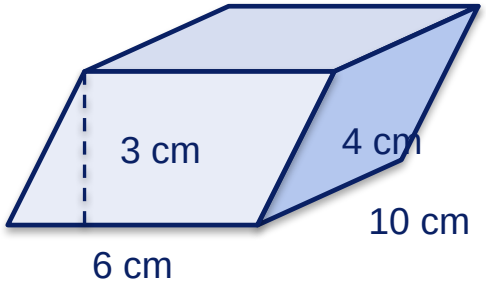
GOLD

Tick or correct in green pen

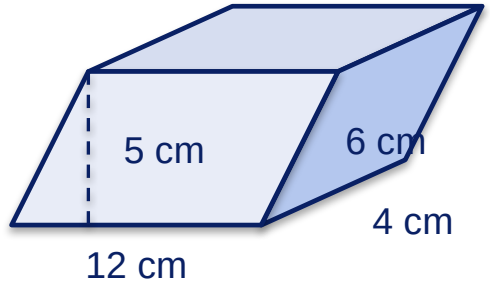
1 **240 cm²**



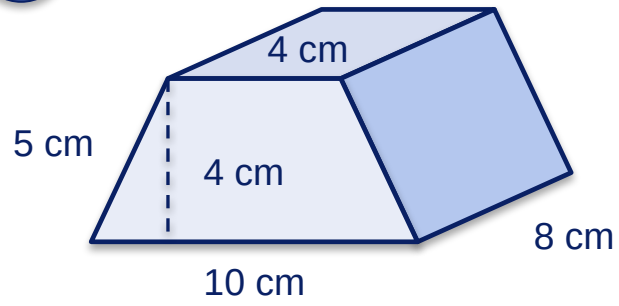
2 **236 cm²**



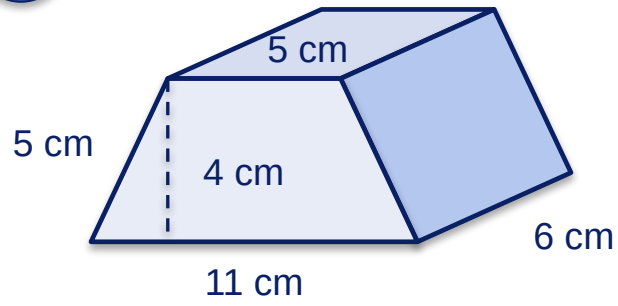
3 **264 cm²**



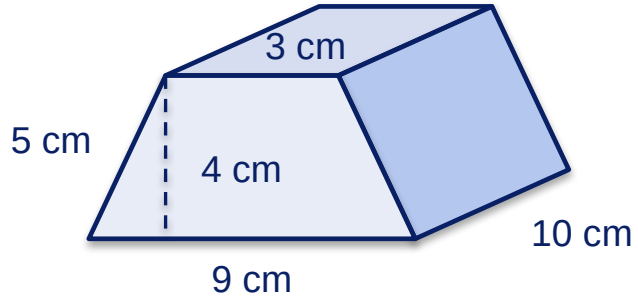
4 **248 cm²**



5 **220 cm²**



★ **268 cm²**



Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Surface area of a cuboid 6 cm by 4 cm by 2 cm

SILVER Triangular prism: sides 3, 4, 5 cm, length 6 cm

GOLD Parallelogram prism: base 6, height 2, slant 3, length 5 (cm)

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **88 cm²**

SILVER **84 cm²**

GOLD **114 cm²**

Exam Practice — Cuboids and Cubes

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

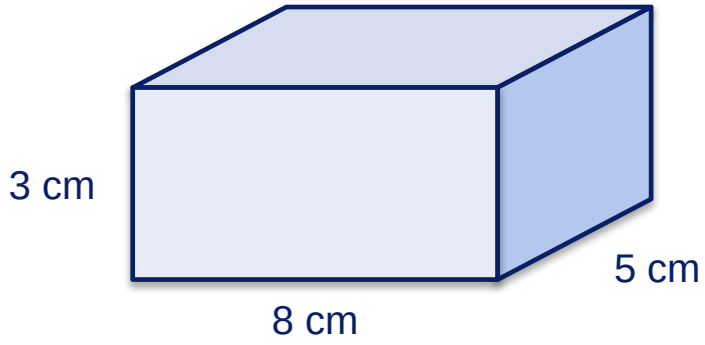
Hint SA = 2(lw + lh + wh) cube: SA = 6 × edge^2

Question 1

(2 marks)

Work out the surface area of this cuboid.

Diagram NOT accurately drawn



..... cm^2

Question 2

(2 marks)

A cube has edges of 7 cm.
Work out its surface area.

..... cm^2

Exam Practice — Triangular Prisms

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

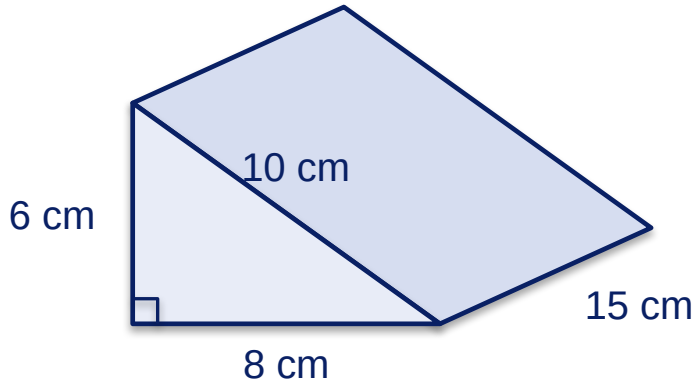
Hint triangle = $\frac{1}{2}$ × base × height

Question 3

(3 marks)

Work out the surface area of this prism.

Diagram NOT accurately drawn



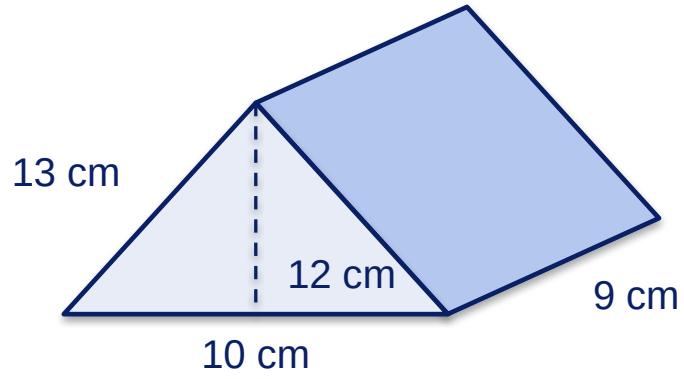
..... cm²

Question 4

(3 marks)

Work out the surface area of this prism.

Diagram NOT accurately drawn



..... cm²

Exam Practice — Other Prisms

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

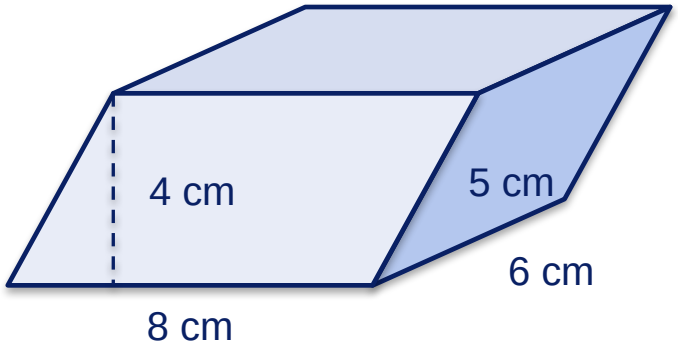
Hint parallelogram = base × height trapezium = $\frac{1}{2}(a + b)h$

Question 5

(3 marks)

The diagram shows a prism.
Calculate its surface area.

Diagram NOT accurately drawn



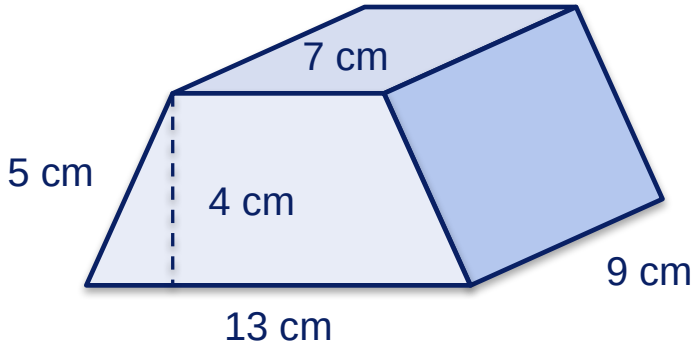
..... cm²

Question 6

(3 marks)

Calculate the surface area
of this prism.

Diagram NOT accurately drawn



..... cm²

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$2 \times (40 + 24 + 15)$ — M1

= 158 cm² — A1

Tip: 8 × 5, 8 × 3 and 3 × 5

Question 2

$6 \times 7 \times 7$ — M1

= 294 cm² — A1

Tip: a cube has 6 equal faces.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$2 \times \frac{1}{2} \times 8 \times 6 = 48$ — M1

$(8 + 6 + 10) \times 15 = 360$ — M1

$48 + 360 = 408\ \text{cm}^2$ — A1

Tip: 2 triangles + 3 rectangles

Question 4

$2 \times \frac{1}{2} \times 10 \times 12 = 120$ — M1

$(10 + 13 + 13) \times 9 = 324$ — M1

$120 + 324 = 444\ \text{cm}^2$ — A1

Tip: the height 12 cm is not an edge.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$2 \times 32 = 64,\ \ 2 \times 48 = 96$ — M1

$2 \times 30 = 60$ — M1

$64 + 96 + 60 = 220\ cm^2$ — A1

Tip: ends 8 × 4, faces 8 × 6 and 5 × 6

Question 6

$2 \times \frac{1}{2} \times (7 + 13) \times 4 = 80$ — M1

$7 \times 9 + 13 \times 9 + 2 \times 5 \times 9 = 270$ — M1

$80 + 270 = 350\ cm^2$ — A1

Tip: the ends are trapeziums.